

Question				Marking details	Marks available														
					AO1	AO2	AO3	Total	Maths	Prac									
		(ii)		award (2) for all three boxes correct award (1) for any two boxes correct <table border="1"><tr><td></td><td>Similarity between hexagonal boron nitride and graphite</td><td>Difference between hexagonal boron nitride and graphite</td></tr><tr><td>Structure</td><td>layers of hexagonal sheets present in both</td><td>layers in register in boron nitride and out of register in graphite / atoms of adjacent layers above one another in boron nitride but not in graphite</td></tr><tr><td>Bonding</td><td>each atom is bonded by covalent bonds to three others in both graphite and BN</td><td>delocalised electrons in graphite but not in boron nitride / bonds are non-polar in graphite but polar in boron nitride</td></tr></table>		Similarity between hexagonal boron nitride and graphite	Difference between hexagonal boron nitride and graphite	Structure	layers of hexagonal sheets present in both	layers in register in boron nitride and out of register in graphite / atoms of adjacent layers above one another in boron nitride but not in graphite	Bonding	each atom is bonded by covalent bonds to three others in both graphite and BN	delocalised electrons in graphite but not in boron nitride / bonds are non-polar in graphite but polar in boron nitride	2			2		
	Similarity between hexagonal boron nitride and graphite	Difference between hexagonal boron nitride and graphite																	
Structure	layers of hexagonal sheets present in both	layers in register in boron nitride and out of register in graphite / atoms of adjacent layers above one another in boron nitride but not in graphite																	
Bonding	each atom is bonded by covalent bonds to three others in both graphite and BN	delocalised electrons in graphite but not in boron nitride / bonds are non-polar in graphite but polar in boron nitride																	
	(c)			phosphorus has available / accessible d-orbitals (1) whilst nitrogen does not (1) or phosphorus can expand its octet (1) but nitrogen cannot (1)	2			2											
				Question 6 total	9	0	0	9	0	0									